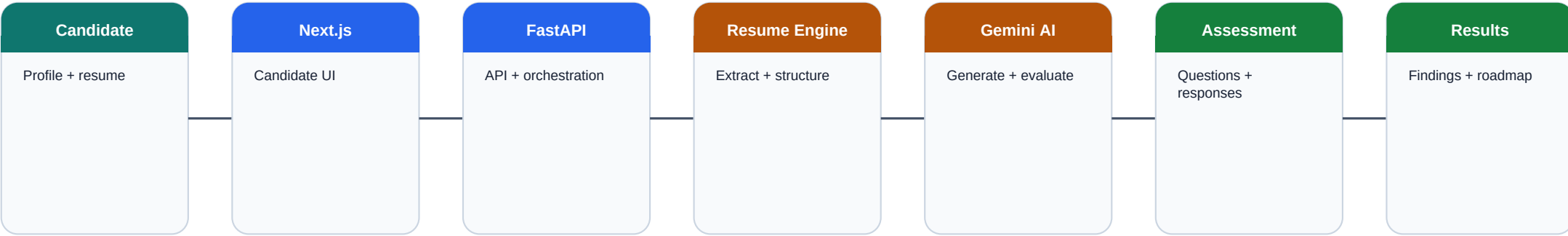


From Resume to Personalized Career Roadmap

A technical flow showing frontend, backend, database, file storage, Gemini AI, assessment and result layers.



MASTER SYSTEM FLOW

Candidate → Next.js → FastAPI → Resume storage/text extraction → AI evidence analysis → personalized questions → assessment responses → evaluation → findings → roadmap → dashboard



Dependency direction: UI → API → Services → Data/Storage/AI

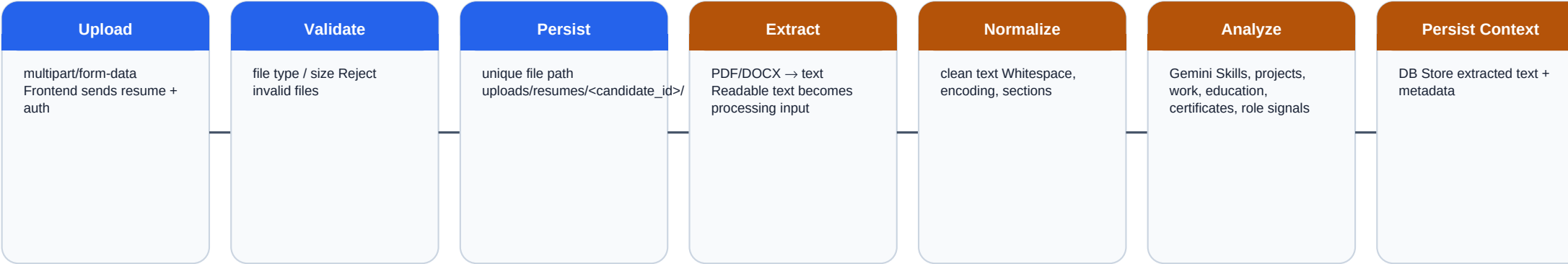
2. Candidate Journey

What the user experiences



3. Resume Processing Pipeline

Original file → structured candidate evidence



EVIDENCE OBJECT

skills[] projects[] experience[] education[] certifications[]
role_signals[]

WHY IT MATTERS

Questions should be grounded in evidence actually found in the resume rather than generic random questions.

FAILURE PATH

Unreadable file → extraction error → clear API error → candidate can replace/re-upload resume.

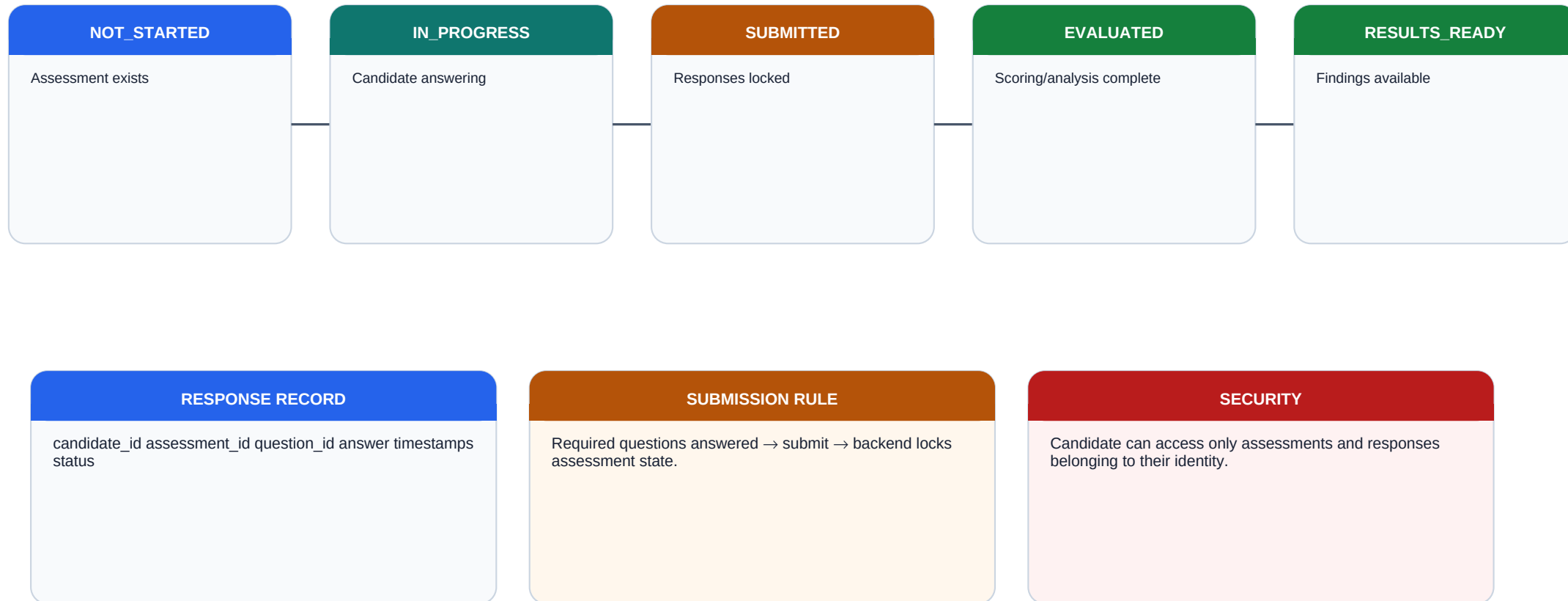
4. Gemini AI Question Generation

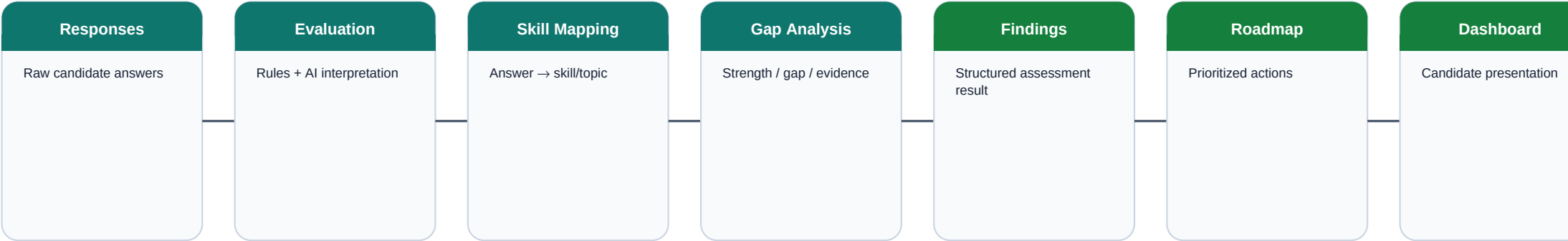
Controlled AI pipeline



5. Assessment Execution & State Machine

Backend remains the source of truth





FINDING OBJECT

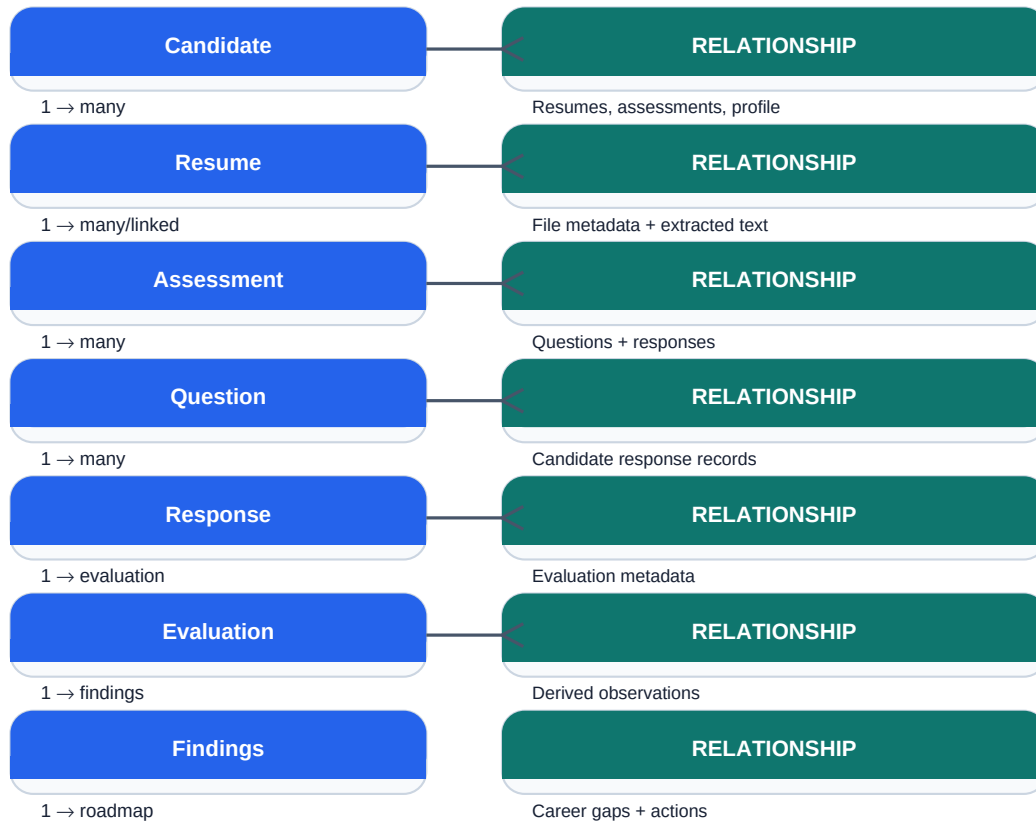
skill status score/observation evidence feedback

ROADMAP OBJECT

priority skill action resource/practice sequence

TRACEABILITY

Every important conclusion should be traceable to questions/responses and/or resume evidence.



FILE STORAGE

uploads/resumes/<candidate_id>/<unique-file> DB stores metadata/reference and extracted text; binary resume remains in file storage.

MIGRATIONS + INTEGRITY

Alembic versions schema. Foreign keys, candidate ownership and validation keep records consistent and isolated.

8. API / Frontend Integration Map

Representative endpoint responsibilities



9. Technology Stack — Where Everything Fits

Current project technology map

Next.js / React / TypeScript	Frontend	Candidate pages, dashboard, assessment UI
FastAPI / Python	Backend	REST API and orchestration
Pydantic	Validation	Request/response contracts
SQLAlchemy	ORM	Database access and models
Alembic	Migration	Versioned schema changes
Relational Database	Persistence	Candidate, resume, assessment and result records
PDF/DOCX tooling	Resume processing	Text extraction
Google Gemini API/SDK	AI	Evidence analysis, questions, evaluation
Local File Storage	Files	Uploaded resumes
Git	Engineering	Version control and submission

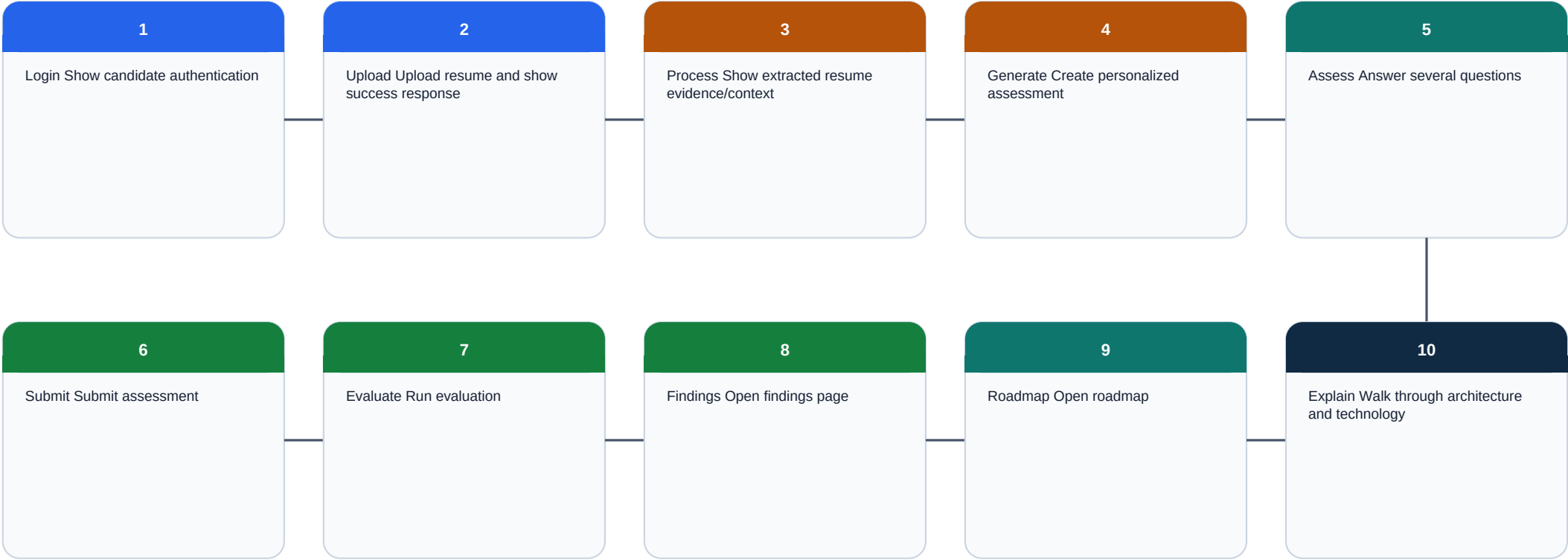
10. Reliability, Security & Failure Paths

What happens when something goes wrong

Auth Failure 401/403 → frontend shows controlled	SYSTEM RESPONSE Never expose protected records	PRINCIPLE Fail safely; never leak secrets or another candidate's data.
Upload Failure Invalid file → reject before processing	SYSTEM RESPONSE Allow replacement/retry	PRINCIPLE Fail safely; never leak secrets or another candidate's data.
Extraction Failure Unreadable document → processing error	SYSTEM RESPONSE Keep original file; request re-upload	PRINCIPLE Fail safely; never leak secrets or another candidate's data.
Gemini Failure Timeout/API/model error → retry or safe fail	SYSTEM RESPONSE Do not persist malformed output	PRINCIPLE Fail safely; never leak secrets or another candidate's data.
DB Failure Transaction/connection error	SYSTEM RESPONSE Return controlled 5xx; preserve consistency	PRINCIPLE Fail safely; never leak secrets or another candidate's data.
Unauthorized Access Ownership check fails	SYSTEM RESPONSE Deny request; no cross-candidate data	PRINCIPLE Fail safely; never leak secrets or another candidate's data.
Frontend Failure API unavailable / malformed response	SYSTEM RESPONSE Show recoverable state and retry action	PRINCIPLE Fail safely; never leak secrets or another candidate's data.

11. Final Demo Flow & Submission Checklist

Recommended order for a hackathon presentation



FINAL MENTAL MODEL

The candidate owns the journey. The frontend owns presentation. FastAPI owns orchestration and security. Services own business logic. DB/storage own persistence. Gemini supplies controlled AI capabilities. Findings and roadmap are derived outputs.